Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/10/2014
PI Name:	O'Banion, Kerry M.D., Ph.D.		
Project Title:	Impact of Space Radiation Induced Neuroinflammation on Alzheimer and Parkinson Disease Pathology		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kerry_obanion@urmc.rochester.edu	**Fax:**	FY 585-756-5334
PI Organization Type:	UNIVERSITY	**Phone:**	585-275-5185
Organization Name:	University of Rochester		
PI Address 1:	Box 603		
PI Address 2:	601 Elmwood Ave		
PI Web Page:			
City:	Rochester	**State:**	NY
Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	11/13/2012	**End Date:**	10/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2016 per S. Monk/LaRC (Ed., 12/11/15)		
Key Personnel Changes/Previous PI:	September 2014: Lisa Opanashuk recently took a position with the FDA and is no longer a University of Rochester Faculty member. We are identifying a new collaborator to assist with analysis of the PD mouse brain sections. September 2013: Sean Hurley was listed in this record as a Co-Investigator; however, Sean left the University prior to the start of this grant and was probably inadvertently included because of the delay between the initial application and the actual funding date. I removed his name from the Co-Investigator section.		
COI Name (Institution):	Olschowka, John (University of Rochester Medical Center) Williams, Jacqueline (University of Rochester Medical Center)		
Grant/Contract No.:	NNX13AC33G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The brain reacts to injury by activation of glial cells and expression of inflammatory mediators in a process called neuroinflammation. Neuroinflammation occurs in the setting of age-associated neurodegenerative diseases such as Alzheimer's (AD) and Parkinson's (PD) and numerous studies suggest a contributory role to disease pathogenesis. We have previously shown that space radiation, at doses approaching those that will be experienced by astronauts on long missions to the moon and Mars, elicits brain neuroinflammation. Therefore we propose to investigate the possibility that space radiation influences pathological changes associated with neurodegenerative diseases. Our studies will utilize three different mouse models of the two most prevalent age-associated neurodegenerative diseases (Alzheimer's and Parkinson's) to assess the effects of space radiation on disease-associated pathology and behavioral changes as well as indices of neuroinflammation and oxidative stress. Findings from these studies will help to refine risk estimates for future space explorers and may offer possible strategies for mitigation of such risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this second year of the grant we are pleased to report significant progress for both Aims 1 and 2. As mentioned in last year's progress report, we arranged for production of sufficient transgenic mice representing both neurodegenerative models (AD and PD) with Jackson Laboratories and in October and November 2013 successfully carried out three separate irradiation campaigns during NSRL-13C. The specific dates and beams were: 1) 300 MeV/μ Silicon at 10 and 100 cGy on October 31; 2) 600 MeV/μ Iron at 10 and 100 cGy on November 15; and 3) 200 cGy mixed protons mimicking a solar flare on November 14. For each ion and energy 10 male and 10 female mice were irradiated. Two sets of sham irradiated (control) mice with 10 males and 10 females in each set were included in the experiment, for a total of 140 transgenic mice. Mice were approximately 7 months of age at the time of irradiation and were shipped back to Rochester to be aged. We have just gathered data related to behavioral analyses, including novel object recognition, open field performance, and contextual fear conditioning for these cohorts, and have just procured tissues from all mice for histopathological analysis of neuroinflammation and Alzheimer's pathology (plaques and tangles) as described in Aim 1 and molecular analyses as described in Aim 3. Notably, the mice were sacrificed at 16-17 months of age, which is a little younger than planned (original plan was 18 months of age), in order to have sufficient number of male and female mice for these studies. Indeed, we had experienced some degree of unexpected death, particularly in males, that does not appear to be correlated to whether they were irradiated or not. Fortunately there should be sufficient mice in all groups to carry out appropriate statistical analyses. Importantly, AD pathology, including evidence of amyloid plaque deposition and hyperphosphorylated tau protein is developed in this mouse model at this age. Thus we should be able to determine whether space radiation at the doses and ions delivered influenced AD pathology or behavioral performance in the tests we conducted. At this time we are currently compiling behavioral data from these many mice and have not yet started the histological analyses, but these will be initiated in October and carried forward into the next funding period. An important aspect of Aim 1 is that it seeks to demonstrate increased AD pathology in a mouse model distinct from the one we used and previously reported showed enhanced amyloid beta plaque accumulation and behavioral deficits following 100 cGy 56Fe particles (1 GeV/μ) (Cherry et al., 2012. PLoS one 7:e53275). With regard to Aim 2, we contracted with Jackson to do a rapid derivation of Parkin-Q311X mice with a goal of 80 mice (40 male, 40 female) harboring the disease model transgene and 80 control littermates. This strategy was instituted because the mice at Jackson are on a mixed background. Importantly, this also provides a non-transgenic control group that will allow us to know whether behavioral effects of radiation interact with genotype. These mice were successfully bred and aged at Jackson Laboratories. They were shipped to BNL in mid April and utilized during the Spring 2014 NSRL run (NSRL14A). The specific dates and beams were: 1) 300 MeV/μ Silicon at 50 cGy on May 1; 2) 600 MeV/μ Iron at 50 cGy on May 12; and 3) 100 cGy mixed protons mimicking a solar flare on May 13. As mentioned in our previous progress report, the cost to obtain sufficient mice to carry out the seven conditions described for Aim 1 was prohibitive; thus the doses used here were selected to provide data for a range of ions at NASA's current "highest" recommended dose for space radiation research. In each instance we irradiated groups of 10 male and 10 female mice representing two genotypes (Parkin-Q311X or wild-type) with each ion and had one set of sham-irradiated control mice. These animals were approximately 8 months of age at the time of irradiation and were shipped to Rochester for further aging. At the current time the mice are out of quarantine, and doing well. We will be starting behavioral tests that are sensitive to basal ganglia dysfunction next month. The specific tests to be used are activity recording, beam walking, screen righting, and tape removal tests and will be repeated at 2 month intervals until sacrifice at 18 months of age. We plan to add an additional set of cognitive tests (novel object recognition and contextual fear conditioning) given singly to sacrifice. Thus tissue will be available for analysis from these mice in early summer of the coming funding period.
Bibliography Type:	Description: (Last Updated: 01/07/2026)
Articles in Peer-reviewed Journals	Sweet TB, Panda N, Hein AM, Das SL, Hurley SD, Olschowka JA, Williams JP, O'Banion MK. "Central nervous system effects of whole-body proton irradiation." Radiat Res. 2014 Jul;182(1):18-34. http://dx.doi.org/10.1667/RR13699.1 ; PubMed PMID: 24937778 , Jul-2014
Articles in Peer-reviewed Journals	O'Banion MK. "Does peripheral inflammation contribute to Alzheimer disease? Evidence from animal models." Neurology. 2014 Aug 5;83(6):480-1. Epub 2014 Jul 2. http://dx.doi.org/10.1212/WNL.0000000000000663 ; PubMed PMID: 24991028 , Aug-2014